HW3: Support Vector Machines (SVMs)

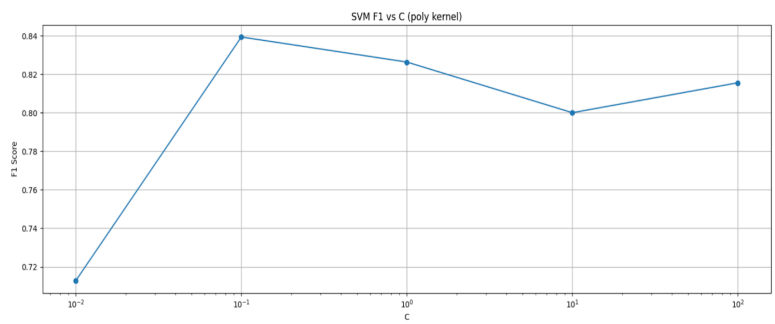
1 Data & Preprocessing

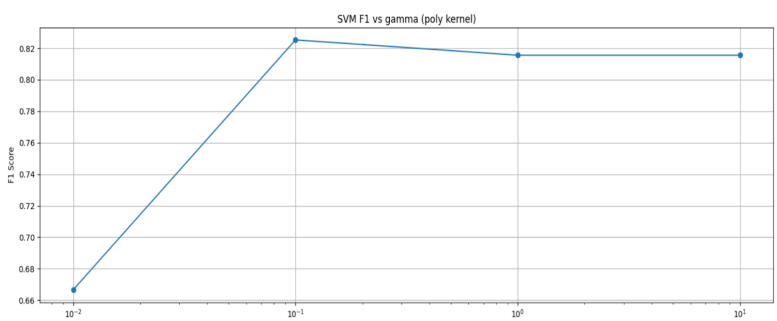
This assignment applied SVMs to two tasks: classification of heart disease and regression for biological aging prediction. The heart disease dataset contains 1025 samples and thirteen features used to predict whether a patient has heart disease. The biological aging dataset consists of DNA methylation data and is very high-dimensional, with 37 samples and over 480 thousand features.

Before training models, preprocessing was performed using leakage-free pipelines. For the heart disease dataset, since there were 723 duplicate rows, the training and testing splits were made so that all duplicate rows were either in one split or the other. Next, for both tasks, missing values were handled using column-wise mean imputation. All features were standardized using StandardScaler to avoid bias caused by larger-magnitude features. Lastly, for the aging dataset, the dimensionality was extremely large relative to the number of samples. Therefore, optional PCA dimensionality reduction was explored to reduce noise and computational complexity before applying SVR models.

2 Hyperparameter Sensitivity

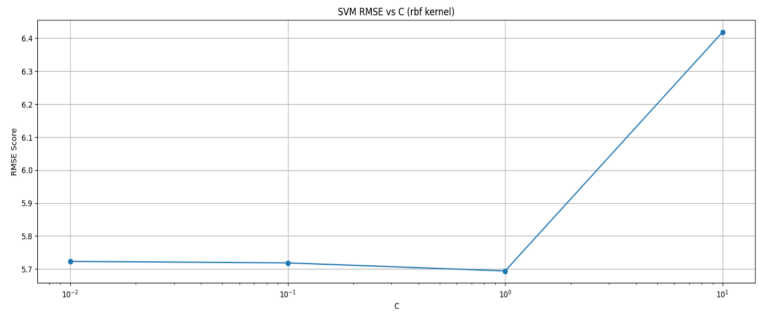
For both classification and regression tasks, the hyperparameters C, gamma, and degree were tuned across three kernels: linear, polynomial, and Radial Basis Function (RBF). The range of hyperparameter values was as follows: C = [0.01, 0.1, 1, 10, 100], gamma = [0.01, 0.1, 1, 10], and degree = [2, 3, 4].

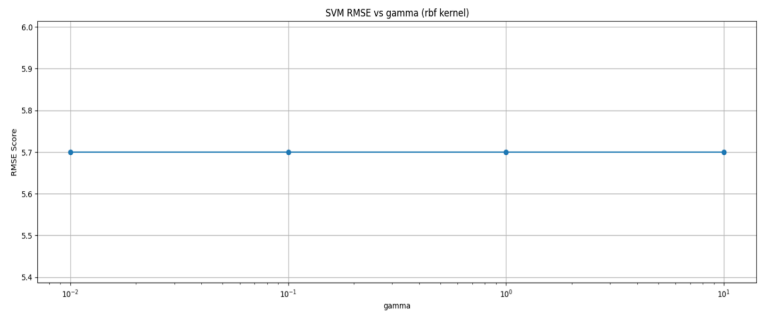




The heart disease SVM classifier performed best on a polynomial kernel with C = 1, gamma = 0.1, and degree = 3, suggesting a nonlinear relationship between features that a linear kernel would not capture. On the other hand, an RBF kernel may be too flexible for the data, causing the model to fit noise rather than a generalizable structure. Therefore, a polynomial classifier of degree 3 can represent cubic relationships between features, which may better approximate the complex relationships underlying heart disease. To the right, we observe varying values of the C and gamma hyperparameters vs. performance for a polynomial kernel. The C hyperparameter controls the tradeoff between margin width and training error. Lower C values result in wider margins and more support vectors, while high C values yield narrower margins and fewer support vectors. Therefore, a more moderate C value of 0.1 or 1 provides a balanced level of regularization that allows the model to maintain a reasonably wide margin while avoiding noise in a relatively small dataset. The gamma hyperparameter controls how strongly individual points influence the decision boundary. Smaller gamma values yield a smoother decision boundary, which could risk underfitting, whereas larger gamma values

yield a more complex decision boundary that risks overfitting. Thus, a more moderate gamma = 0.1 allowed the polynomial kernel to capture nonlinear relationships without creating overly complex boundaries.





The age prediction SVM regression model achieved its best performance using an RBF kernel with PCA dimensionality reduction, with hyperparameters C = 1 and gamma = 0.01. Applying PCA reduces the dimensionality by projecting the data into a smaller set of orthogonal components that capture the largest variance in the data. The RBF kernel allows the model to capture complex nonlinear relationships between genomic features and biological age, especially with such a high-dimensional dataset. Similar to the classification model, a value of C = 1 provides moderate regularization, allowing the model to learn meaningful patterns without fitting noise. From the plot on the right, we observe that a larger C value of 10 yields a worse RMSE score, as the model attempts to fit the training data more closely, increasing model complexity and leading to overfitting, which worsens generalization error on the test set. On the other hand, hyperparameter tuning on gamma yielded similar results, indicating that the SVR model was relatively insensitive to this parameter within the tested range. This likely occurs because PCA significantly reduced the dimensionality of the data and produced a smoother feature space, making the RBF kernel behave similarly for multiple gamma values. Additionally, the small sample size and moderate regularization (C = 1) limited the flexibility of the regression function, resulting in nearly identical predictions across gamma values.

3 Metrics

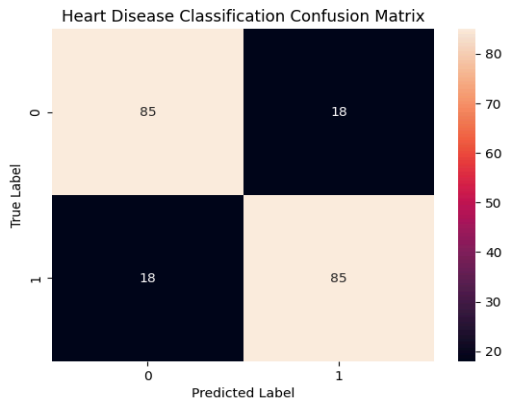
The best heart disease SVM classifier resulted in the following metrics:

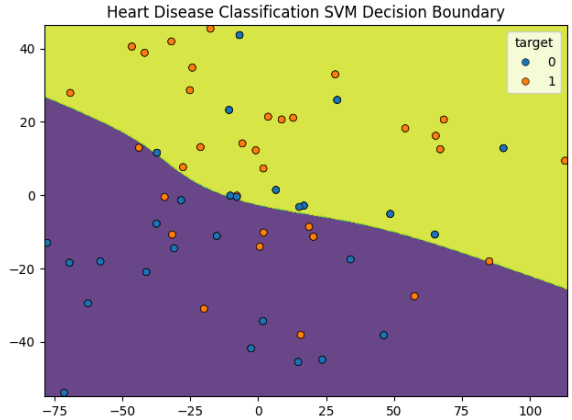
	Accuracy	Precision	Recall	F1	ROC-AUC
Training	0.977	0.957	1.0	0.978	0.999
Testing	0.825	0.825	0.825	0.825	0.869

We observe fairly high testing metrics, indicating that the model performs fairly well on new data. However, these values are lower than the training metrics, which suggests that the model does not generalize perfectly to unseen data. This drop in performance is consistent with moderate overfitting as seen by the very high training metrics, which is expected given the relatively small dataset.

From the confusion matrix, the model produces equal numbers of false positives and false negatives, indicating a relatively balanced classification error rate. However, fewer false negatives than false

positives would be preferable for heart disease classification, as observed in the training metrics, since we want to avoid cases where a patient is predicted not to have heart disease when they actually do. Similarly, we observe some incorrect predictions in the decision boundary plot below.

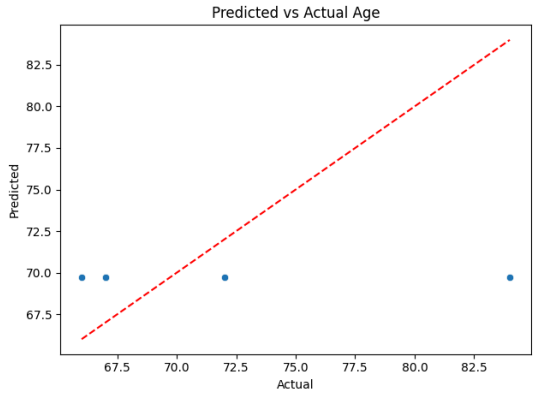


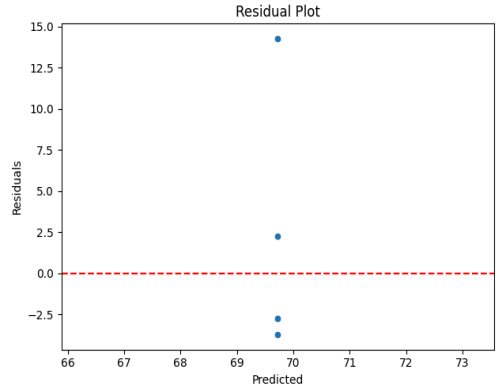


The best age prediction SVM regressor resulted in the following metrics:

	MAE	RMSE	R2
Training	5.489	8.443	-0.090
Testing	4.237	5.700	-0.0003

The near-zero R2 score indicates that the model performs roughly as well as predicting the mean age. However, although this RBF SVR does not fit the training data very well, its test performance is better and more stable than that of other kernel and hyperparameter configurations that exhibited extreme overfitting. Additionally, from the parity and residual plots, we see that the model predicts the same age for all 8 test samples. All of these behaviors likely occur due to the very small sample size of 37, which makes the data prone to overfitting and does not provide enough information to adequately train the models.





4 Discussion

Several limitations of both the data and modeling approaches likely contributed to the results described. First, the aging dataset contains far more features than samples, which makes it easy for models to overfit and difficult to identify reliable predictive signals. Even with PCA, the very limited number of samples limits the model's ability to generalize. Second, the heart disease dataset contained a large number of duplicate rows. While these rows were not removed, these duplicates significantly limited the diversity of training examples. Third, SVM models are sensitive to hyperparameter selection and feature scaling, and their performance can vary substantially depending on these choices, as indicated by the hyperparameter exploration outlined.